4 PM BATCH – Questions, Answers

Question 1 – Data Types

What will be the output of the following Python code?

```
x = 10
y = 3.0
z = x / y
print(type(z).__name__)
```

A. int

B. float

C. double

D. number

Correct Option: B. float

Reason: Division using / produces a float value in Python.

Question 2 – Data Types

What will be the output of the following Python code?

```
x = 10
y = x
x = 20
print(y)
```

A. 10

B. 20

C. 30

D. Error

Correct Option: A. 10

Reason: y keeps the value 10 even after x is reassigned.

Question 3 – Data Types

What will be the output of the following Python code?

```
x = 10
```

```
y = 10.0
print(x == y)
print(type(x) == type(y))
```

A. True / True

B. False / False

C. True / False

D. False / True

Correct Option: C. True / False

Reason: The values are equal, but their data types are different.

Question 4 – Data Structures

What will be the output?

```
numbers = [10, 20, 30, 40, 50]
print(numbers[1:4])
```

A. [10, 20, 30]

B. [20, 30, 40]

C. [20, 30, 40, 50]

D. [10, 20, 30, 40]

Correct Option: B. [20, 30, 40]

Reason: The slice includes indices 1 through 3.

Question 5 – Data Structures

What will be the output?

```
data = [10, 20, 30]
data.append([40, 50])
data.extend([60, 70])
print(data)
```

A. [10, 20, 30, 40, 50, 60, 70]

B. [10, 20, 30, [40, 50], 60, 70]

C. [10, 20, 30, [40, 50], [60, 70]]

D. [10, 20, 30, 40, 50, [60, 70]]

Correct Option: B. [10, 20, 30, [40, 50], 60, 70]

Reason: append adds one element, while extend adds elements individually.

Question 6 – Data Structures

What will be the output?

```
data = {'A': 10, 'B': 20, 'C': 30}
data['B'] = data['A'] + data['C']
print(data['B'])
```

A. 20

B. 30

C. 40

D. 50

Correct Option: C. 40

Reason: B is updated to 10 + 30.

Question 7 – Data Structures

What will be the output?

```
items = [1, 2, 2, 3, 3, 3]
unique_items = set(items)
result = tuple(unique_items)
print(len(result))
```

A. 3

B. 4

C. 5

D. 6

Correct Option: A. 3

Reason: A set removes duplicate values.

Question 8 – Data Structures

What will be the output?

```
text = 'Data Science'
result = text[::2]
print(result)
```

A. Dt cec

B. DtaSine

C. DtSine

D. aaSine

Correct Option: A. Dt cec

Reason: Every second character is selected starting from index 0.

Question 9 – Data Structures

What will be the output?

```
numbers = [10, 20, 30, 20, 40]
print(numbers.count(20))
```

A. 1

B. 2

C. 3

D. 4

Correct Option: B. 2

Reason: count returns the number of occurrences.

Question 10 – Data Structures

What will be the output?

```
student = {'name': 'Rahul', 'age': 22, 'course': 'Data Science'}
result = student.get('marks', 75)
print(result)
```

A. None

B. 75

C. marks

D. KeyError

Correct Option: B. 75

Reason: get returns the default value when the key is absent.

Question 11 – Conditional Statements

What will be the output?

```
x = 15
y = 10
if x > y:
    if x % 2 == 0:
        result = 'Even'
    else:
        result = 'Odd'
else:
    result = 'Smaller'
print(result)
```

A. Even

B. Odd

C. Smaller

D. 15

Correct Option: B. Odd

Reason: 15 is greater than 10 but is not even.

Question 12 – Conditional Statements

What will be the output?

```
x = 12
y = 8
z = 15
if x > y and z > x:
    result = 'A'
elif x > y or z < y:
    result = 'B'
else:
    result = 'C'
print(result)
```

A. A

B. B

C. C

D. Error

Correct Option: A. A

Reason: Both conditions in the first if statement are true.

Question 13 – Conditional Statements / Recursion

What will be the output?

```
def calculate(n):
    if n <= 1:
        return 1
    return n + calculate(n - 2)

print(calculate(5))
```

A. 9

B. 10

C. 11

D. 15

Correct Option: A. 9

Reason: The recursive calls calculate 5 + 3 + 1.

Question 14 – Conditional Statements

What will be the output?

```
num = 121
reverse = 0
temp = num
while temp > 0:
    digit = temp % 10
    reverse = reverse * 10 + digit
    temp //= 10
if num == reverse:
    result = 'Palindrome'
else:
    result = 'Not Palindrome'
print(result)
```

A. Palindrome

B. Not Palindrome

C. 121

D. Error

Correct Option: A. Palindrome

Reason: The reversed number equals the original number.

Question 15 – Conditional Statements

What will be the output?

```
num = 29
is_prime = True
if num < 2:
    is_prime = False
else:
    for i in range(2, int(num ** 0.5) + 1):
        if num % i == 0:
            is_prime = False
            break
print('Prime' if is_prime else 'Not Prime')
```

A. Prime

B. Not Prime

C. True

D. Error

Correct Option: A. Prime

Reason: 29 has no divisor other than 1 and itself.

Question 16 – Loops

What will be the output?

```
total = 0
for i in range(1, 6):
    if i % 2 == 0:
        continue
    total += i
print(total)
```

A. 6

B. 9

C. 15

D. 10

Correct Option: B. 9

Reason: Only 1, 3 and 5 are added.

Question 17 – Loops

What will be the output?

```
n = 5
result = 1
while n > 1:
    result *= n
    n -= 2
print(result)
```

A. 15

B. 20

C. 30

D. 60

Correct Option: A. 15

Reason: The loop multiplies 5 × 3.

Question 18 – Loops / Lambda

What will be the output?

```
words = ['python', 'data', 'science', 'ai']
result = []
for word in words:
    f = lambda x: x.upper() if len(x) > 4 else x
    result.append(f(word))
print(result)
```

A. ['PYTHON', 'DATA', 'SCIENCE', 'AI']

B. ['PYTHON', 'data', 'SCIENCE', 'ai']

C. ['python', 'data', 'SCIENCE', 'AI']

D. ['PYTHON', 'DATA', 'science', 'AI']

Correct Option: B. ['PYTHON', 'data', 'SCIENCE', 'ai']

Reason: Only words longer than four characters are uppercased.

Question 19 – Loops / Strings

What will be the output?

```
text = 'PYTHON'
result = ''
for i, ch in enumerate(text):
    if i % 2 == 0:
        result += ch.lower()
    else:
        result += ch
print(result)
```

A. pYtHoN

B. PyThOn

C. pythOn

D. PytHoN

Correct Option: A. pYtHoN

Reason: Characters at even indices are converted to lowercase.

Question 20 – Loops

What will be the output?

```
number = 12345
total = 0
while number > 0:
    digit = number % 10
    if digit % 2 != 0:
        total += digit
    number //= 10
print(total)
```

A. 9

B. 15

C. 6

D. 25

Correct Option: A. 9

Reason: Only odd digits 5, 3 and 1 are added.

Question 21 – Loops / Dictionary

What will be the output?

```
marks = {'Python': 78, 'SQL': 85, 'Statistics': 62, 'Machine Learning': 91}
count = 0
for subject, mark in marks.items():
    if mark >= 80:
        count += 1
    elif mark >= 60:
        count += 0.5
print(count)
```

A. 2

B. 2.5

C. 3

D. 3.5

Correct Option: C. 3

Reason: Two scores add 1 each and two scores add 0.5 each.

Question 22 – Loops / Coding Round

What will be the output?

```
n = 6
result = 0
for i in range(1, n + 1):
    if n % i == 0:
        result += i
print(result)
```

A. 10

B. 12

C. 14

D. 16

Correct Option: B. 12

Reason: The divisors 1, 2, 3 and 6 sum to 12.

Question 23 – NumPy

What will be the output?

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
result = np.sum(arr, axis=0)
print(result)
```

A. [6 15]

B. [5 7 9]

C. [1 2 3]

D. [4 5 6]

Correct Option: B. [5 7 9]

Reason: axis=0 performs a column-wise sum.

Question 24 – NumPy

What will be the output?

```
import numpy as np
a = np.ones((2, 3))
b = np.zeros((2, 3))
result = a + b
print(result)
```

A. [[1. 1. 1.] [1. 1. 1.]]

B. [[0. 0. 0.] [0. 0. 0.]]

C. [[1 1 1] [1 1 1]]

D. [[2. 2. 2.] [2. 2. 2.]]

Correct Option: A. [[1. 1. 1.] [1. 1. 1.]]

Reason: Adding zeros to ones keeps every value equal to 1.0.

Question 25 – NumPy

What will be the output?

```
import numpy as np
arr = np.arange(1, 13)
result = arr.reshape(3, 4)
print(result[1:, 1:3])
```

A. [[6 7] [10 11]]

B. [[5 6] [9 10]]

C. [[6 7 8] [10 11 12]]

D. [[2 3] [6 7]]

Correct Option: A. [[6 7] [10 11]]

Reason: The slice selects rows from index 1 and columns 1 to 2.